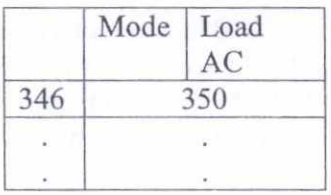
Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science &Engineering

B. E.(COE/CSE) : EST	**Course Code: UCS510/UCS319** **Course Name: Computer Architecture and Organization**
4/Dec/2024	**2:00 PM – 5:00PM**
Time: 3 Hours, M. Marks: 40	**Faculty: SHI/SOL/AAS/YAS/ESH**

Note: ***All questions carry equal marks. Attempt any 4 questions.***

Attempt questions in sequence and assume for any missing data.

Q.No	Questions	MM	CO	Level
Q1.A	Consider a 16-bit processor where the following contents appear in the main memory for a two-word instruction, starting at location 345, as shown in the figure. The first part of the instruction indicates the mode field specifies the addressing mode, and the second part loads a value into an accumulator. A source register is R1, which has a value of 538. There is also a base register that contains the value 250. The value 350 in location 346 may be part of the address calculation. Assume that location 1099 contains a value of 1500, location 1098 contains a value of 1499, 1097 contains a value of 1498 and so on. Determine the effective address and operand to be loaded in the accumulator for the following address modes: immediate, register Indirect, auto-decrement, indirect relative and base register. (all numbers are in decimal)  	6	CO2	L3
Q1.B	Explain any four logical key difference between the followings: (i) RISC vs CISC processors. (ii) Memory Mapped vs I/O Mapped.	2+2	CO3	L2
Q2. A	Consider a five stage pipelining with duration 150, 100, 160, 180 and 80 ps. The latch delay between each stage is 5, 10, 9, 3 and 8 ps. Compute the followings: (i) Speed up Ratio (ii) What would be the total time required to process 500 data items on pipelining and non-pipeline processor. (iii) Throughput (iv) If 4th stage (i.e. 180) of given pipeline is further divided into two stage with 110 and 70 ps. What is the speedup ratio achieved with the new configuration?	6	CO3	L5
Q2.B	Consider a processor with memory unit of having 8192 words and each word of size 32 bit. Explain the special purpose registers namely PC, AR, DR, TR, IR, AC, INPR and OUTR along with their size for the given configuration in detail.	4	CO2	L4

Q3. A	Consider the sequence of machine instructions given below: MUL R5, R0, R1 DIV R6, R2, R3 ADD R7, R5, R6 SUB R8, R7, R4 In the above sequence, R0 to R8 are general purpose registers. This sequence of instructions is to be executed in a pipelined instruction processor with the 4 stages: Instruction Fetch and Decode (IF), Operand Fetch (OF), Perform Operation (PO) and Write back the result (WB). The IF, OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD or SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses operand forwarding from the PO stage to the OF stage. Draw a space time diagram that represents instruction flow through the pipeline during each clock cycle without operand forwarding and with operand forwarding. How many clock cycles are required to execute the above sequence of the instructions?	5	CO3	L4
Q3. B	What are different types of pipeline hazards. Explain each of them with a suitable example. How these hazards can be resolved/ prevented? Provide different solutions for each type of hazard.	5	CO3	L2
Q4. A	A block-set associative cache memory consists of 256 blocks divided into four block sets. The main memory consists of 32,768 blocks and each block contains 256 eight bit words. (i) How many bits are required for addressing the main memory? (ii) How many bits are needed to represent the TAG, SET and WORD fields? (iii) What is the size of tag directory?	4	CO4	L5
Q4.B	(i) A direct mapped cache memory of 2MB has a block size of 512 bytes. The cache has an access time of 3 ns and a hit rate of 96%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. What would be the average memory access time? (ii) Suppose, there are two level of cache. Access time of Level 1, Level 2 and main memory are 2ns, 20ns and 600ns respectively. The hit rates of Level 1 and Level 2 caches are 70% and 80% respectively, what is the average access time seen by the processor ignoring the search time within the cache?	3+3	CO4	L5
Q5.A	Explain the step by step working of DMA when memory needs to transfer files of size 50MB to I/O. Explain each involved signal, register and pins. What are the different mode of transfer used by DMA? Explain each of them.	6	CO4	L4
Q5.B	Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through *cycle stealing* at regular intervals. Consider a 2 MHz processor, if 0.5% processor cycles are used for DMA. What is the data transfer rate of the device? (in bits per second).	4	CO4	L5